Post-hoc Calibration Under Covariate Shift: An Analysis of Classical Tabular Models

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Abstract

Temperature scaling and related post-hoc maps are often treated as a cheap fix for miscalibrated probabilities. We measure whether that fix *transfers* when test covariates shift, for logistic regression, random forests, and histogram gradient boosting on small tabular datasets (no deep nets, no API models). On a Gaussian mean-shift of four features at strength 1.5, the difference $\Delta\text{ECE} = \text{ECE}_{\text{shift}} - \text{ECE}_{\text{iid}}$ is positive in 51/60cells without calibration (mean 0.113±0.149) and remains positive after i.i.d. temperature, isotonic, or histogram calibration (56, 53, and 57of 60cells; one-sided Wilcoxon p=2.2e-10). The effect is **not** universal: histogram boosting on breast-cancer has $\Delta\text{ECE} = 0.002 \pm 0.009$. Post-hoc maps can help or hurt relative to doing nothing (wine random forests: isotonic changes shifted ECE by -0.123; breast-cancer logistic regression: 0.074). Unweighted split-conformal coverage falls with shift strength (breast-cancer logistic regression: 0.913 at s=0 to 0.778 at s=1.5). We report all seeds, failed hypotheses, and shift-family controls. This is an analysis result, not a new calibrator.

1 Introduction

A classifier is *calibrated* when a predicted probability p is correct with frequency p. Post-hoc maps—Platt-style scaling, histogram binning, isotonic regression, and temperature scaling—are the standard way to improve calibration without retraining [1–3]. A separate literature shows that deep-net uncertainty *collapses* under dataset shift [4–6]. Almost all of that evidence is vision-scale neural nets. Tree ensembles remain strong on tabular data [7], and their i.i.d. calibration properties have been known for decades [2], but we found no verified paper that measures whether i.i.d.-fitted post-hoc maps still work for these models under an explicit covariate shift of $P(X)$.

We treat that as a measurement question. We do not introduce a new calibrator. A negative or mixed answer is an acceptable outcome.

Primary hypothesis (H1). Fitting temperature scaling, isotonic regression, or histogram binning on an i.i.d. validation split does not keep ECE from rising when test features are mean-shifted.

Two other hypotheses were piloted and *not* selected as the paper claim: isotonic overfit vs. temperature as a function of n_{cal} (H2; heterogeneous), and oracle-weighted conformal recovery (H3; the 1-D weight estimator failed). Decision log: `logs/decisions.md`.

2 Related work

Surveys of calibration methods and metrics include [8, 9]. ECE via confidence binning [3, 10] is biased and definition-dependent [11–13]; we therefore also report Brier score [14] and NLL [15]. Temperature scaling [3] and Dirichlet calibration [16] target neural nets. Unsupervised recalibration under covariate shift exists for neural models [17]. Domain-drift post-hoc calibration of CNNs is studied by [6]. We reuse those *questions*, not their model class.

Conformal prediction has finite-sample coverage under exchangeability [18–20] and requires density-ratio weights under covariate shift [21]. We use unweighted split conformal as a secondary diagnostic, not as a proposed fix.

3 Methods

Models. Scikit-learn logistic regression (with standard scaling), random forests (100 trees), and `HistGradientBoostingClassifier`. All expose `predict_proba`.

Calibrators. Fitted on a held-out calibration split of the *source* distribution, then frozen. `none`: identity. `temperature`: one scalar T on $\log p$, fitted by NLL (Guo-style, using log-probabilities so trees without logits still apply). `isotonic`: sklearn isotonic regression on $p(y{=}1)$ (binary) or on confidence (multiclass). `histogram`: equal-width binning of the same 1-D score.

Shift. Primary: add $s \cdot \hat{\sigma}_j$ to features $j \in \{0, 1, 2, 3\}$ of the test set only (training and calibration unshifted). Strengths $s \in \{0, 1.0, 1.5, 2.5\}$. Ablations: quantile slice and exponential-tilt resampling on feature 0; a control that instead shifts the *last* four columns.

Metrics. 15-bin equal-width confidence ECE, Brier, NLL, accuracy, and split conformal coverage with APS scores $1-p_y$ at $\alpha = 0.1$.

Protocol. Each dataset is split 50% train / 25% calibrate / 25% test, stratified, with seeds $\{0, 1, 2, 3, 4\}$. Results are written only by `calibshift.io.write_result` to `results/*.json`. Paper macros in `paper/numbers.tex` are generated from those files.

4 Experiments

Datasets. Offline-only: sklearn `breast_cancer` (binary) and `wine` (3-class); two `make_classification` sets (binary and 3-class). No OpenML, no paid APIs.

Main grid (`exp04_main_h1`). 240 cells, 0 failed, 23.0 seconds on CPU. This is the source of all headline numbers.

Ablations. `exp05`: quantile slice and importance resampling. `exp06`: $n_{\mathrm{cal}} \in \{50, 100, 200\}$ (killed primary H2). `exp07`: shift trailing columns.

Sanity. I.i.d. logistic regression on breast-cancer: accuracy 0.979, ECE 0.032±0.010(linear models on this set are typically $\gtrsim$ 0.95 accurate). Wine random forests are overconfident i.i.d. (ECE 0.147±0.020), matching the classical tree-calibration picture [2]. Split conformal coverage at s=0 is near the nominal $1 - \alpha = 0.9$ for α=0.1. We do *not* compare ECE magnitudes to ImageNet temperature scaling [3].

5 Results

5.1 H1: ECE usually rises under mean-shift

Figure 1 plots shifted ECE against s. Figure 2 shows mean ΔECE at s=1.5. Pooled over 60 dataset×model×seed cells, uncalibrated ΔECE = 0.113 ± 0.149 (51 positive, 9 negative; Wilcoxon p=2.2e-10). After i.i.d. temperature scaling the pooled mean is 0.124±0.150(p=3.1e-11). Isotonic and histogram are similar (0.141 and 0.141). Synthetic logistic regression is the largest effect (0.292±0.214) and also the noisiest; breast-cancer histogram boosting is the exception (0.002±0.009, interval includes 0). Table 1 records the pooled statistics.

5.2 Post-hoc maps are not a reliable remedy

Figure 3 plots $\mathrm{ECE}_{\mathrm{shift}}(\mathrm{cal}) - \mathrm{ECE}_{\mathrm{shift}}(\mathrm{none})$. Wine random forests *benefit* from isotonic (-0.123 ECE; shifted ECE 0.175 → temperature 0.068), because the uncalibrated forest was already badly calibrated. Breast-cancer logistic regression is *hurt* by isotonic (0.074). We therefore do not rank calibrators under shift.

Table 1: Pooled ΔECE at s=1.5 (n = 60 cells: 4 datasets × 3 models × 5 seeds). Source: `results/stats_tests.json`, `summary_main.json`.

Calibrator	mean ΔECE	std	#pos/neg	Wilcoxon p
none	0.113	0.149	51/9	2.2e-10
temperature	0.124	0.150	56/4	3.1e-11
isotonic	0.141	0.151	53/7	4.6e-11
histogram	0.141	0.151	57/3	3.3e-11

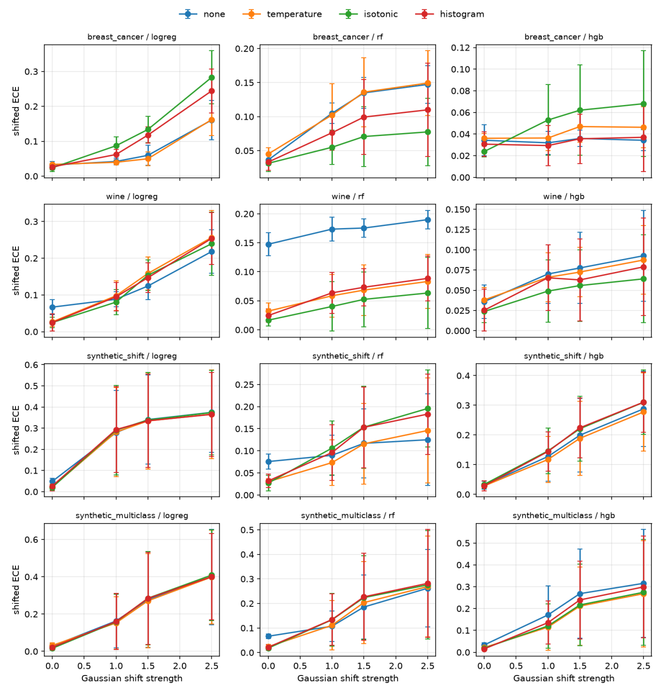

Figure 1: Shifted ECE vs. Gaussian mean-shift strength (mean ± sd over 5 seeds). s=0 is i.i.d. Source: `results/exp04_main_h1.json`.

5.3 Coverage tracks broken exchangeability

Unweighted split conformal (logreg, `none`) coverage at α = 0.1: breast-cancer 0.913 (s=0), 0.778 (s=1.5), 0.608 (s=2.5); synthetic binary 0.634 at s=1.5 (Figure 4). This is predicted by theory [21]; we are not claiming a new conformal method. A 1-D “oracle” density ratio on feature 0 did *not* restore coverage in the H3 pilot (killed).

5.4 Controls

Quantile slicing induces small ΔECE relative to the Gaussian mean-shift (exp05 JSON). Importance resampling can *decrease* ECE on breast-cancer (the resampled test is not harder). Shifting trailing columns on `synthetic_shift` yields RF ΔECE = 0.009 vs. 0.041 when shifting the first four columns: trees are less sensitive to unused coordinates; logistic regression is not. H2 (isotonic vs. temperature vs. n_{cal}) remains mixed and is not a paper claim.

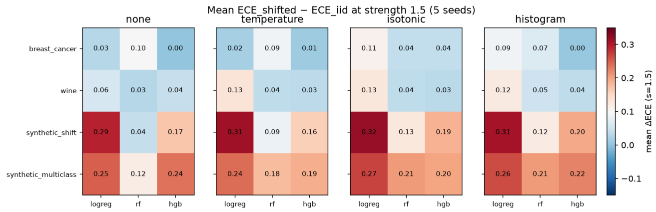

Figure 2: Mean ΔECE at s=1.5. Positive (red) means the shifted test is *worse* calibrated than the i.i.d. test.

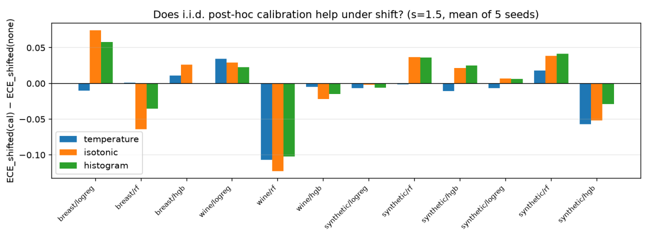

Figure 3: Does i.i.d. calibration help on the shifted test? Negative bars mean the calibrator beats `none` under shift.

6 Limitations

- **Shift is synthetic.** A four-feature mean-shift is not WILDS [22] and is entangled with accuracy drop (synthetic logreg accuracy 0.932→0.620 at s=1.5).
- **Small tabular data only.** No ImageNet, no language models.
- **Multiclass isotonic/histogram** map *confidence*, not a full Dirichlet map [16].
- **ECE estimator.** 15 equal-width bins; we also logged quantile-binned ECE in the JSON but did not promote it to a claim.
- **Temperature on probabilities.** Trees have no logits; we scale $\log p$. That is not identical to logit-temperature on a neural net.
- **H3 weights were misspecified.** A 1-D histogram ratio is not $p_{\text{test}}(x)/p_{\text{cal}}(x)$ for a 4-D shift.
- **Pooling.** The headline Wilcoxon pools heterogeneous dataset×model cells. Per-dataset tests are in `results/stats_tests.json`; we do not hide the HGB exception.

7 Conclusion

For sklearn tabular models, i.i.d. post-hoc calibration is not a dependable answer to covariate shift of the kind we applied. ECE usually rises, conformal coverage falls, and the popular maps can make shifted ECE better *or* worse than doing nothing. The honest exception (breast-cancer HGB) and the failed H2/H3 hypotheses are part of the result.

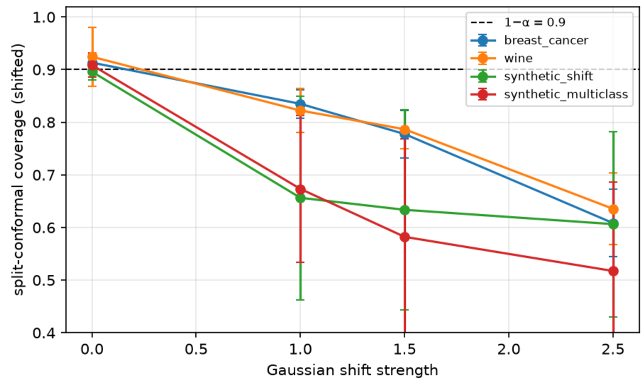

Figure 4: Unweighted split-conformal coverage vs. shift (logreg, no calibrator). Dashed line: $1 - \alpha = 0.9$.

AI-generation disclosure

This repository and draft were produced by a coding agent executing the user's `GOAL.md` protocol (literature verification via arXiv and Semantic Scholar APIs, experiments on CPU, no hand-edited `results/*.json`). Every numeric macro in `paper/numbers.tex` is generated from those files. Citations are restricted to `paper/verified.bib`. A human should still read the limitations and re-run `make fresh-clone-test` before treating the PDF as camera-ready.

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